Data Analytics and Machine Learning for solar power generation

ENERGY DATA ANALYTICS

Paolo Finardi

Energy & Finance Institute

www.energyandfinanceinstitute.com



Paolo Finardi

Head of
Data Science & AI

- **Leads AI Strategy** and a team of 30+ Data Scientists across the Retail, E-Mobility, and Renewables sectors.
- **10+ years of experience** focused on scalable ML solutions in Forecasting and Energy Optimization.
- **Active community contributor**: Member of the Databricks Product Advisory Board and Organizer of PyData Milan.

www.linkedin.com/in/paolo-finardi-38b25096

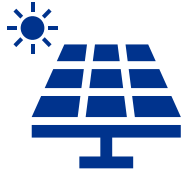
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Data science in PV: three families of use cases



Predict

How much will we produce in the next hours and days?

Output: Production forecast

Used by: Trading, grid scheduling



Diagnose

Is the plant producing what it should? Where not, and why?

Output: KPIs, alarms, work orders

Used by: O&M, asset management



Optimize

When should storage charge and discharge?

Output: Dispatch schedule

Used by: Trading, asset management

Predict: forecasting PV production

The question

How much will the plant produce in the next minutes, hours or days?

Output

A production profile per time interval, ideally with its uncertainty

Used by

Trading desk, grid scheduling, plant operations

Data

- **Historical production:** SCADA power and energy
- **Weather forecasts:** irradiance, cloud cover, temperature from numerical weather models
- **Measured weather:** on-site irradiance and temperatures
- **Satellite and sky images:** for very short horizons
- **Plant metadata:** location, capacity, tilt, orientation

Algorithms

Baselines
Persistence, recent averages

Physical models
Weather forecast → irradiance on panels → power

Statistical & ML
ARIMA, Prophet, regression, gradient boosting, random forest

Deep learning
LSTM, transformers; often combined in hybrids

Diagnose: finding where a plant loses energy

The question

Is the plant producing what it should? If not, where and why?

Output

Performance KPIs, alarms and a prioritised list of interventions

Used by

O&M teams, asset managers

Data

- **SCADA:** inverter and string power, currents, voltages
- **Weather station:** irradiance on the panels, module temperature
- **Alarms and maintenance logs:** events, interventions
- **Topology:** layout, ratings, which strings go to which inverter
- **Inspections:** thermography, drone images

Algorithms

Rules & KPIs
Thresholds, performance ratio, availability

Expected-power models
Physics or regression on irradiance and temperature; analyse the gap

Anomaly detection
Isolation Forest, autoencoders, one-class SVM

Root-cause models
Classifiers on labelled failures, computer vision on images

Optimize: deciding when to store and when to sell

The question

When should a battery next to the plant charge and discharge?

Output

A charge and discharge schedule per time interval

Used by

Trading desk, asset managers

Data

- **Production forecasts:** from the Predict use case
- **Price forecasts and market data:** day-ahead and intraday
- **Battery state:** state of charge, power limits, efficiency, degradation
- **Constraints:** grid connection limit, contracts, tariffs

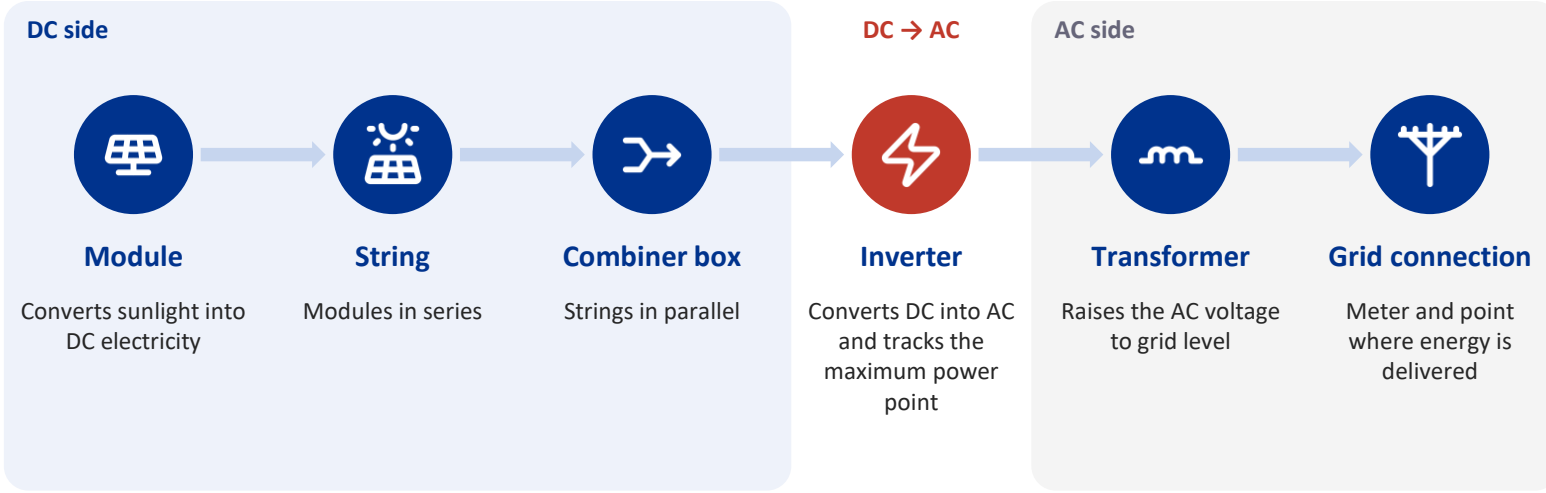
Algorithms

Rules
Charge when prices are low, discharge at peaks

Linear & mixed-integer programming
LP / MILP with Pyomo

Reinforcement learning
Learns a dispatch policy by trial and error

PV 101: how a plant works, from module to grid



Weather station on site
Irradiance and temperatures: the inputs that drive production

What can be measured at each step

Step	What can be measured	How common
Weather station	Irradiance (horizontal and on the panel plane), ambient and module temperature, wind speed	Standard in utility-scale plants
Module	Rarely monitored continuously; checked with thermography, drone images, I-V curves	Inspections; per-module data only with module electronics
String	String current, sometimes voltage	Common in large plants
Combiner box	Currents per string, DC voltage, fuse and surge-protection status	Common in large plants
Inverter	DC and AC power, voltage, current, frequency, internal temperature, status and alarm codes, energy counters	Always
Transformer	Temperatures, breaker status, power on the grid side	Usually, at substation level
Grid connection	Revenue meter (energy delivered), grid operator setpoints and curtailment orders	Always, often in separate systems

Our dataset: two solar plants in India

34 days
15 May – 17 June 2020

15 minutes
one reading per interval

22 inverters
per plant

4 CSV files
two per plant

File	Level	One row is…	Rows
Plant_1_Generation_Data.csv	Inverter	one inverter, one 15-min interval	68,778
Plant_1_Weather_Sensor_Data.csv	Plant	the weather sensor, one 15-min interval	3,182
Plant_2_Generation_Data.csv	Inverter	one inverter, one 15-min interval	67,698
Plant_2_Weather_Sensor_Data.csv	Plant	the weather sensor, one 15-min interval	3,259

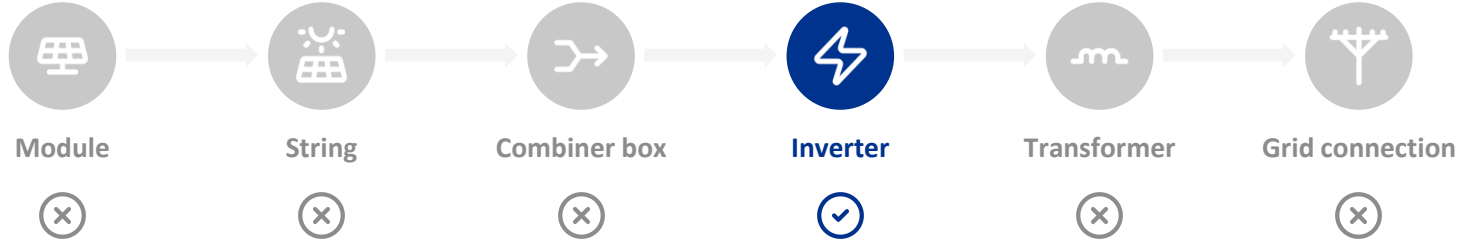
Source: Kaggle — Solar Power Generation Data (anikannal) · a public sample dataset, ideal for learning the method

Data dictionary: what each column means

Column	File	Meaning	Unit
DATE_TIME	Both	Start of the 15-minute interval	timestamp
PLANT_ID	Both	Plant identifier	—
SOURCE_KEY	Both	Inverter ID (generation) or sensor ID (weather)	—
DC_POWER	Generation	DC power entering the inverter	kW
AC_POWER	Generation	AC power leaving the inverter	kW
DAILY_YIELD	Generation	Energy produced since midnight (cumulative)	kWh
TOTAL_YIELD	Generation	Energy produced since the inverter started (cumulative)	kWh
IRRADIATION	Weather	Solar irradiation measured by the plant sensor	kW/m^2
AMBIENT_TEMPERATURE	Weather	Air temperature	°C
MODULE_TEMPERATURE	Weather	Temperature measured on a module	°C

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What our dataset sees, and what it does not



Weather sensor (plant level)
IRRADIATION, AMBIENT_TEMPERATURE, MODULE_TEMPERATURE

We see the plant through its inverters. Problems upstream (modules, strings) or downstream (transformer, grid) can only be inferred from how inverters behave. No alarms, no logs, no capacity data.

First look at a new dataset: questions to ask

Structure and time

What is one row?
The grain: timestamp × inverter? And the keys to join files

Is time clean?
Format, time zone, regular frequency, gaps, duplicates

Is everything there?
Expected rows vs received rows, per device and per day

Which units?
kW or W, kWh or MWh: check the order of magnitude

Values and consistency

Does night look like night?
Production and irradiance at zero after sunset

Any impossible values?
Negative power, values above capacity, frozen readings

Do related measures agree?
Power vs energy counters, DC vs AC, power vs irradiance

Can I see it?
Plot one device on one day before any aggregate

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Physical checks every PV dataset must pass

Check	What physics says	Red flag in the data
Night	No sun, no production	Power or irradiance at night: time zone or timestamp problem
Sun and power	Power rises and falls with irradiance	Peaks out of sync: time shift; large scatter: losses or bad data
DC vs AC	AC is slightly lower than DC (inverter losses)	AC above DC, or a ratio far from one: units or scaling error
Capacity	Power never exceeds what the plant can deliver	Values above capacity: wrong units or wrong metadata
Temperatures	Under the sun, modules are hotter than the air	Module colder than air at noon: sensor swapped or misplaced
Power vs energy	Energy = Σ power × time; counters never decrease within a day	Mismatch: one of the two is wrong; drops: resets or replacements

Common problems along the chain

Level	Typical problem	Signature in the data
Module	Soiling, degradation, cracks, hot spots	Slow decline of output vs irradiance; soiling recovers after rain or cleaning
String	Disconnected string, blown fuse	Step drop in the inverter's DC power; string current at zero
Combiner box	Fuse or connector failure	Larger step drop: several strings lost at once
Inverter	Trip or outage	Zero output while the sun shines and neighbours produce
Inverter	Clipping, thermal derating	Flat ceiling at the inverter rating, lower on hot days
Transformer	Fault or maintenance	All inverters behind it stop at the same moment
Grid	Curtailment, grid outage, voltage trips	Plant-wide cap or drop; whole plant at zero; trips at peak hours

Data problems look like plant problems

Communication loss
Missing rows, or zeros and frozen values while the plant was producing

Wrong units or scale
A variable off by a constant factor (W vs kW, ×10)

Time shifts
Peaks out of sync with the sun; one-hour jumps at daylight-saving changes

Sensor issues
Dirty or misaligned irradiance sensor, drifting temperature probe

Counter resets and duplicates
Energy counters dropping, the same reading recorded twice

Changing identifiers
A replaced inverter appears under a new ID

Before blaming the plant, double-check the data.

Hands-on

Exploring the data from scratch

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Notebooks for the hands-on

Github Repo

01_data_exploration_plant1.ipynb

02_data_exploration_plant2.ipynb

03_plant2_from_diagnosis_to_work_order.ipynb

04_data_modelling_plant2_forecasting.ipynb

Forecast horizons: how far ahead, with which data

Hands-on #2



Minutes

Data

Sky cameras, latest measurements

Used for

Real-time plant and grid control

Hours

Data

Satellite images, recent production

Used for

Intraday scheduling

Day ahead

Data

Weather forecasts, production history

Used for

Next-day scheduling and market bids

Days to weeks

Data

Weather forecasts, seasonal averages

Used for

Maintenance planning

The further ahead, the more the forecast depends on weather forecasts, and the larger the uncertainty.

Time series: order matters

Cross-sectional data

Many units at the same moment, e.g. all inverters at 12:00.

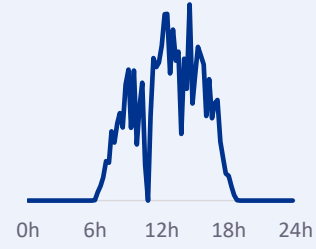
Rows are independent: their order carries no information.

Inverter	Power
A	x_1
B	x_2
C	x_3
D	x_4

Time series

One unit at successive moments, e.g. the plant every 15 minutes.

Each value depends on the previous ones: the order is the information.



What changes for the data scientist

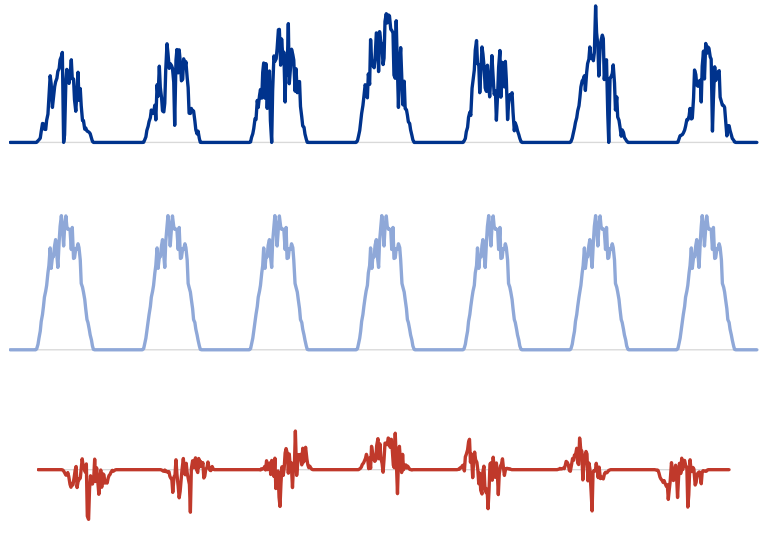
- **Never shuffle the rows**
- **Split train and test by time**
- **The past is a feature: yesterday, last week**

A PV series = trend + daily cycle + weather

Observed

Daily cycle

Weather & noise



Trend
Slow changes over months: seasons, ageing of the panels. Flat within a week.

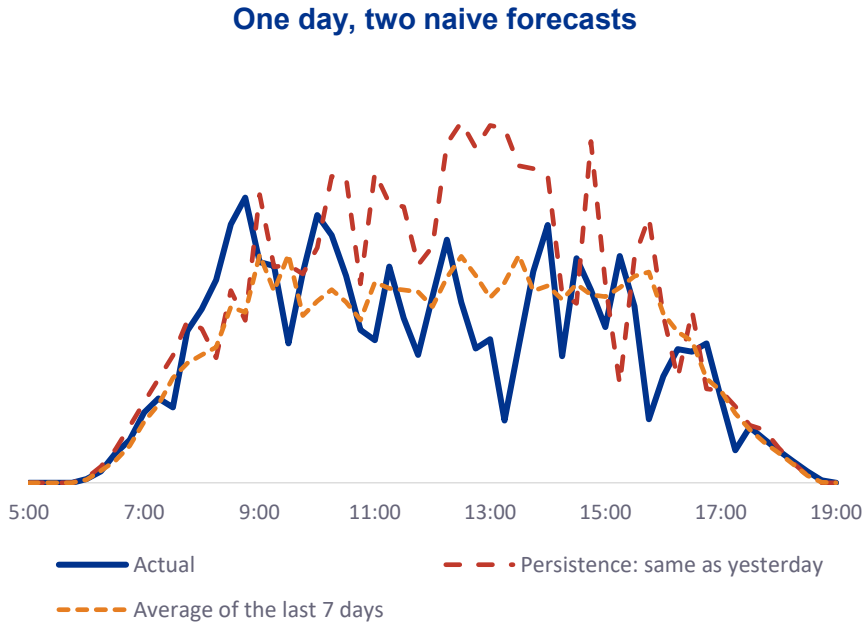
Daily cycle
The sun's path: the same shape every day. Easy to learn.

Weather & noise
Clouds and everything else. This is what a forecast really has to predict.

Additive or multiplicative? In PV the daily cycle scales with the weather: a cloudy day is a smaller copy of a sunny day.

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Baselines first: the bar every model must clear



Persistence

Tomorrow will be like today.
Strong when the weather is stable, wrong at every change.

Average profile

Tomorrow will be like the recent average day.
Smooth and robust, but blind to tomorrow's weather.

A model is useful only if it beats both.

Prophet in one slide

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t)$$

g(t) · Trend

Slow changes, modelled as straight segments joined at changepoints.

Knob: changepoint_prior_scale: how freely the trend bends

s(t) · Seasonality

Repeating cycles (daily, weekly, yearly), built from sums of sine waves.

Knob: fourier_order: how detailed the cycle is; additive or multiplicative mode

h(t) · Events & regressors

Holidays and external drivers, such as irradiance.

Knob: add_regressor(): tell the model what it cannot see in its own past

Strengths: readable components, robust to gaps and outliers, fast to try.

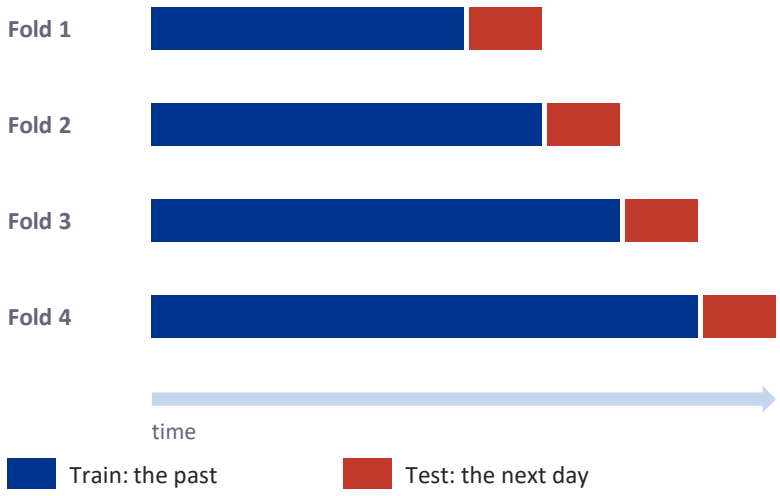
Watch out: from its own past alone it learns the average day, not tomorrow's clouds.

Learn more: explain-me-like-im-5-how-facebook-prophet-works-and-how-to-tune-it

Taylor & Letham (2018), Forecasting at Scale, The American Statistician · Prophet documentation

Evaluate like in production

Rolling-origin backtest



The model never sees the future it is tested on, exactly as in real operations.

MAE — **average of |actual – forecast|**
The typical error, in kW.

RMSE — **√ average of (actual – forecast)²**
Punishes large misses more.

nMAE — **MAE ÷ plant capacity**
Comparable across plants of different size.

Beware of MAPE. It divides by the actual value: at dawn, dusk and night production is close to zero and the error explodes. Evaluate on daylight hours.

Hands-on | Forecasting & Backtesting

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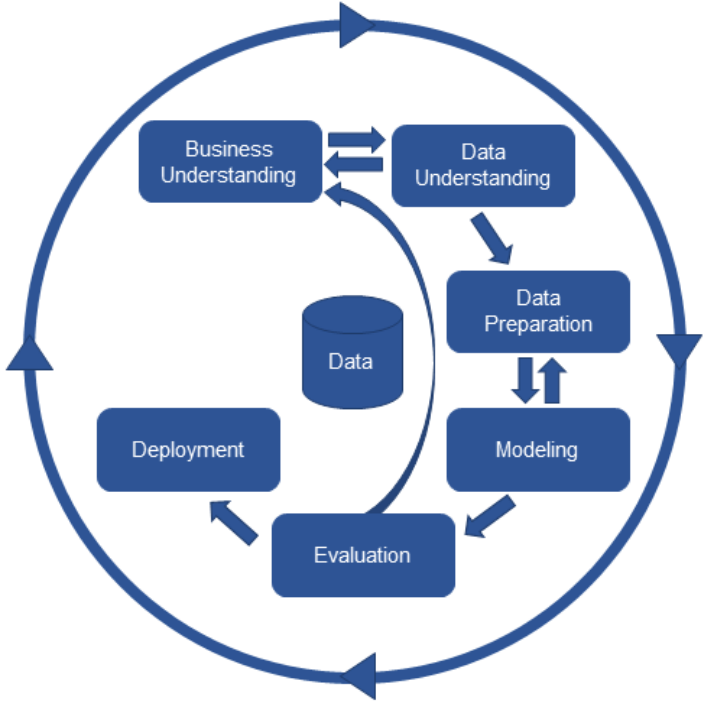
Q & A

Appendix

Deep dives

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Data Science Project Lifecycle



Business Understanding

Define project objectives and understand business requirements to guide data science efforts.

Data Understanding

Collect and explore data to identify quality, patterns, and initial insights.

Data Preparation

Clean and transform data to create a suitable dataset for modeling and analysis.

Modelling

Build models using algorithms to extract patterns and make predictions.

Evaluation

Assess model performance using metrics to ensure accuracy and reliability.

Deployment & Business Translation

Implement the model in production and translate results into actionable business insights.

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1) Business Understanding



Phase-Specific Questions
What problem are we solving?
What metric defines success?

Tangible Deliverables
Business Canvas

Start with Business Questions

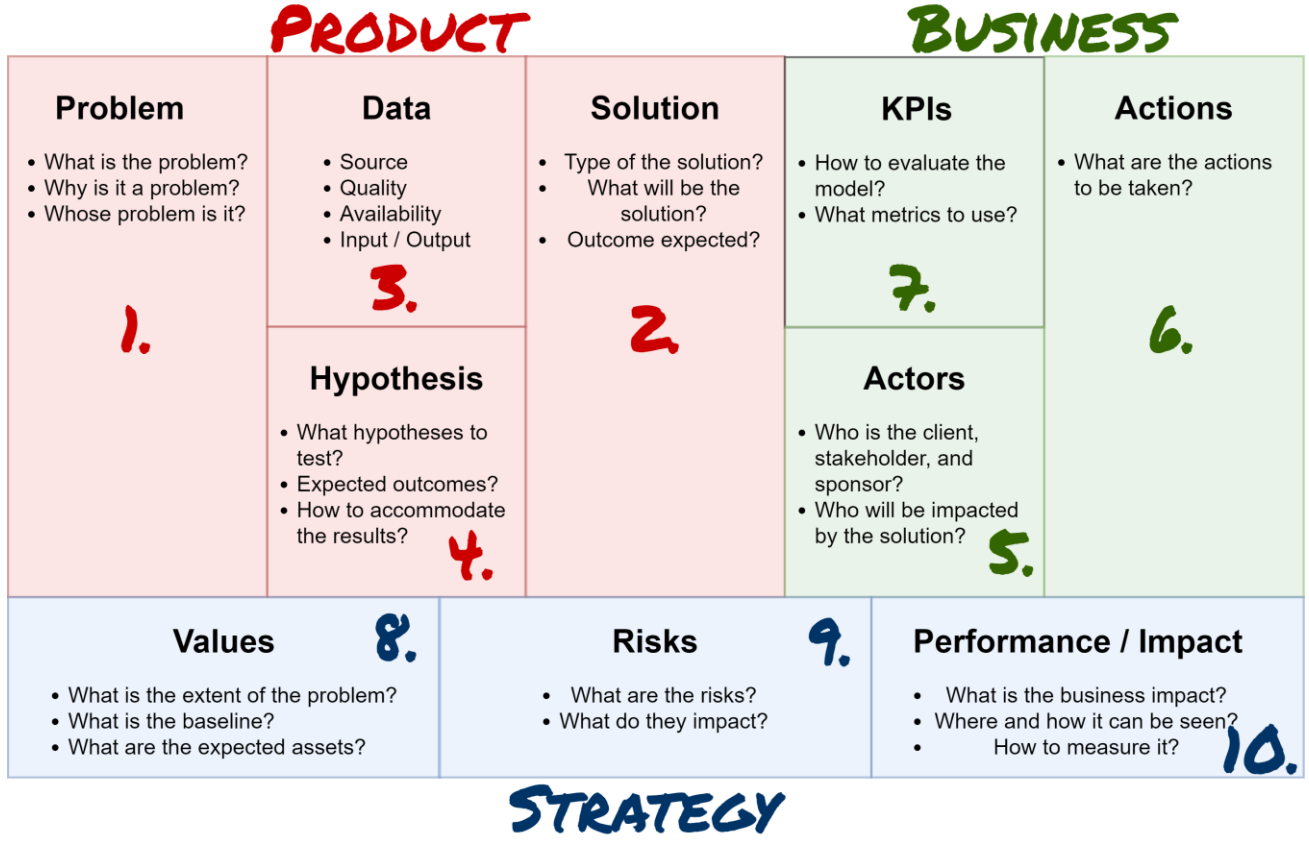
Begin by framing clear business questions that focus on measurable impact and actionable outcomes.

Define Success Metrics

Set clear success metrics such as error targets (e.g. nMAE), cost savings, and environmental impact for effective evaluation.

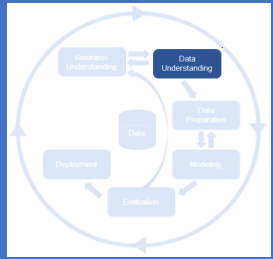
Identify Stakeholders and Timeline

Recognize key stakeholders, decisions affected, and the time horizon for implementing solutions.



https://www.acronymat.com/2023/08/22/data-model-canvas/

2) Data Understanding



Phase-Specific Questions
What data do we have?
Is it reliable?

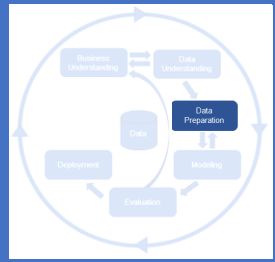
Tangible Deliverables
Data inventory,
initial exploration

Data Collection & Inventory
Identify all available data sources and their structure (type, frequency, format, variables).

Data Quality Assessment
Evaluate completeness, consistency, and accuracy of the datasets.

Initial Exploration
Perform preliminary analyses to uncover distributions, correlations, and potential anomalies.

3) Data Preparation



Phase-Specific Questions
How do we clean and structure the dataset?

Tangible Deliverables
Ready-to-model dataset (Master Table)

Feature Set Creation

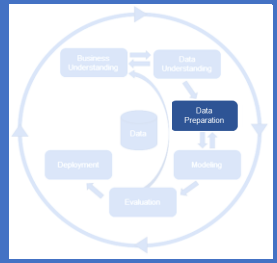
Lag Features
Lag features use past time points like P t-1 and P t-24 to capture temporal dependencies in data.

Rolling Statistics
Rolling means or standard deviations smooth data to reveal trends and variability over time.

Calendar and Weather Features
Features include hour, weekday, season and weather indices like clear-sky irradiance for richer input data.

Cyclic and Interaction Features
Cyclic encodings like sin/cos for time and interactions between variables enhance feature informativeness.

3) Data Preparation



Phase-Specific Questions
How do we clean and structure the dataset?

Tangible Deliverables
Ready-to-model dataset (Master Table)

Handling Missing Values
Identify and manage missing data to ensure dataset completeness and accuracy for modeling.

Correcting Outliers
Detect and adjust outliers to improve data quality and model reliability.

Feature Set Creation
Relevant features are extracted from training data to create the final feature set for modeling.

4) Data Modelling



Phase-Specific Questions
Which algorithms or approaches?

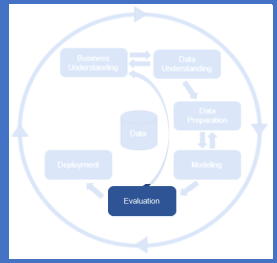
Tangible Deliverables
Predictive models object

Start from simple models
Simple baseline models, like persistence, provide foundational benchmarks for forecasting tasks.

Importance of Baseline Comparison
Compare new methods to a well-established baseline to ensure meaningful improvements.

Use Experiment books
Track models improvements with an experiment book to avoid rolling dice modeling

5) Evaluation



Phase-Specific Questions
Performance match the expectation? Is it stable?

Tangible Deliverables
Metrics, validation results

Validation Techniques
Apply backtesting or rolling window validation to respect time order and simulate real operations.

Business-Relevant Metrics
Use MAE, RMSE and nMAE (MAE / installed capacity). MAPE breaks down in PV: production is zero at night.

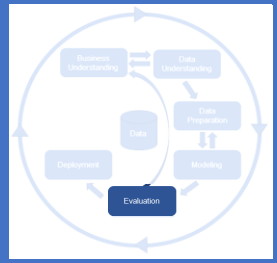
Linking Metrics to Business Impact
Translate forecast error into €: imbalance costs, lost revenue, O&M priorities.

5) Evaluation

Evaluation Metrics for time series

Metric	Formula	What it Measures	Pros	Cons
MAE (Mean Absolute Error)	$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	Average magnitude of errors between predictions and actuals, without considering direction.	• Easy to interpret (same units as target). • Robust to outliers compared to RMSE.	- Treats all errors equally, doesn't penalize large deviations.
RMSE (Root Mean Squared Error)	$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	Square root of the average of squared differences between predicted and actual values.	• Highlights large errors, good when big mistakes are costly. • Easy to interpret (same units as target).	• Sensitive to outliers, can overemphasize few large errors.
MAPE (Mean Absolute Percentage Error)	$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$	Average absolute percentage difference between predicted and actual values.	• Scale-independent, allows comparison across sites or systems. • Easy to communicate as percentage accuracy.	• Undefined at zero: explodes at dawn, dusk and night in PV. Prefer nMAE (MAE / capacity)

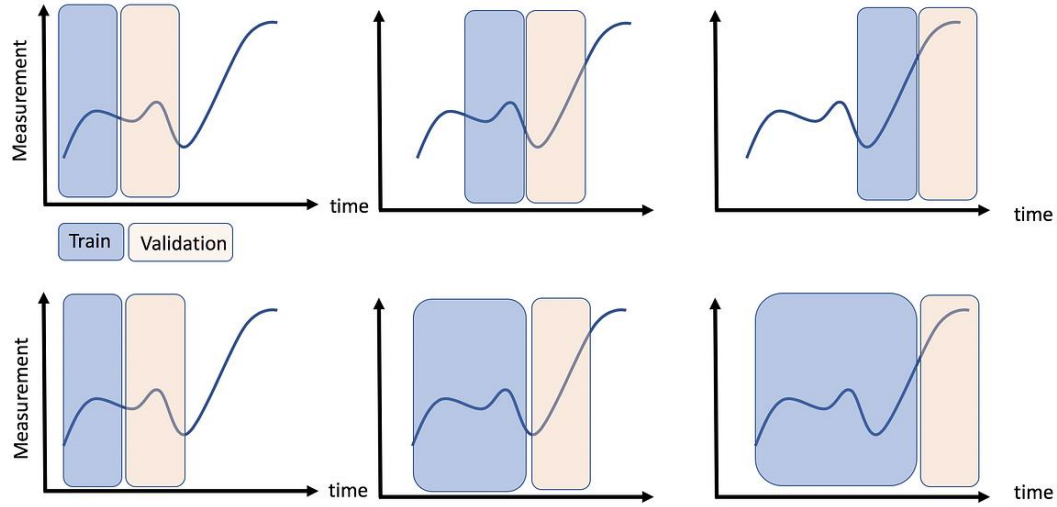
5) Evaluation



Phase-Specific Questions
Performance match the expectation? Is it stable?

Tangible Deliverables
Metrics, validation results

Validation Techniques
Apply backtesting or rolling window validation to respect time order and simulate real operations.



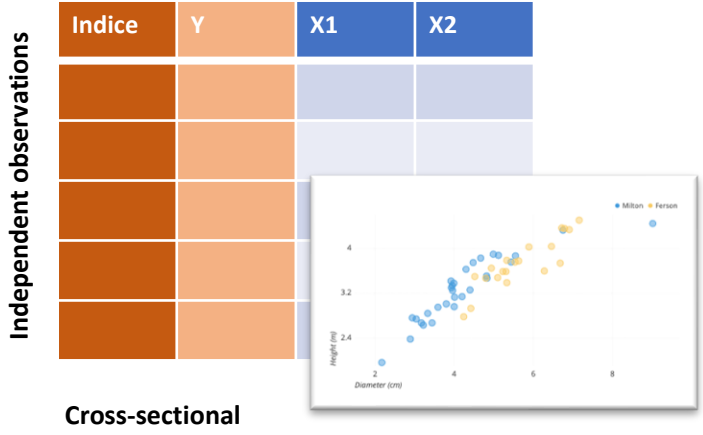
Iterative nature and deliverables of each phase

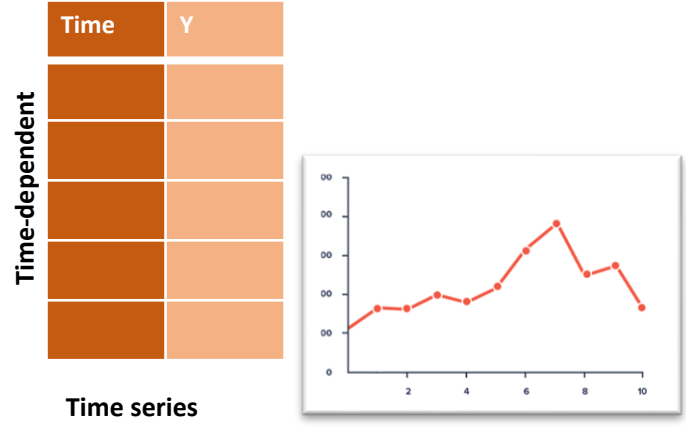
Phase	Key Questions	Deliverables
1. Business Understanding	What problem are we solving? What metric defines success?	Business Canvas
2. Data Understanding	What data do we have? Is it reliable?	Data inventory, initial exploration
3. Data Preparation	How do we clean and structure it?	Ready-to-model dataset
4. Modelling	Which algorithms or approaches?	Predictive models
5. Evaluation	Does it work? Is it usable?	Metrics, validation results
6. Deployment / Translation	How do we deliver the value?	Dashboard, decision support, automation

What is a time series

Cross-sectional data: observations at a fixed point in time or where time does not matter.

Time series: Datasets sorted and observed at successive points in time





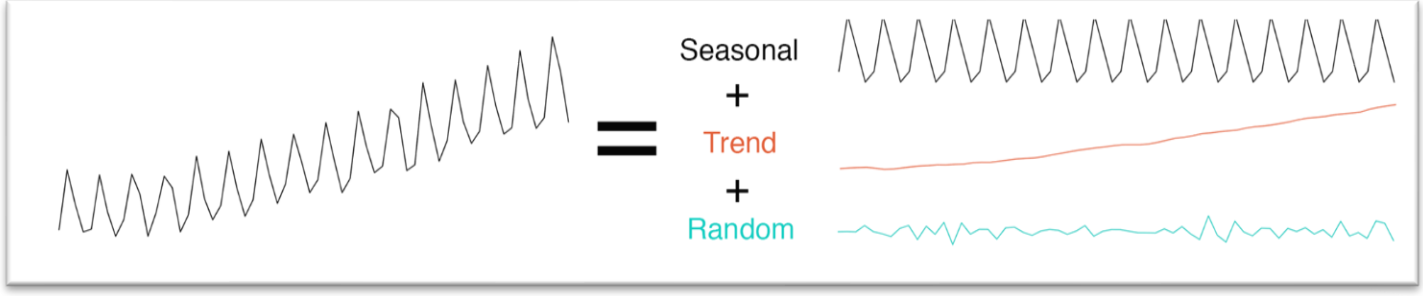
Time series decomposition

To make time series forecasts the simplest way is **decomposition**
In fact, each time series can be seen as the sum of a signal plus a noise

$$Time\ series = Signal + Noise$$

In the same way, the signal can be decomposed into different components such as trend and seasonality

- Multiplicative models
  $Signal = trend * stagionalità * noise$
- Additive Models
  $Signal = trend + stagionalità + noise$



Facebook's Prophet

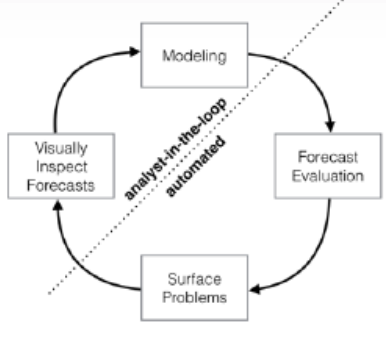
"Forecasting at Scale" di Taylor e Letham (2017)

Generalized Additive Model (GAM) with the aim of encouraging the use of domain knowledge within the forecasting model

Additive Model:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

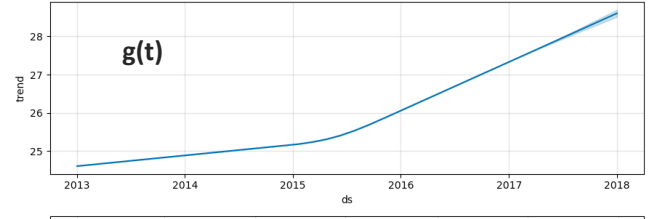
Regression on **trend** g(t), **seasonality** s(t), **special events** h(t)

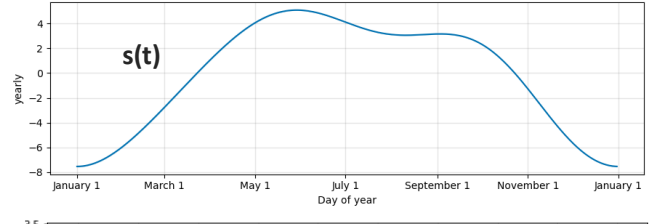


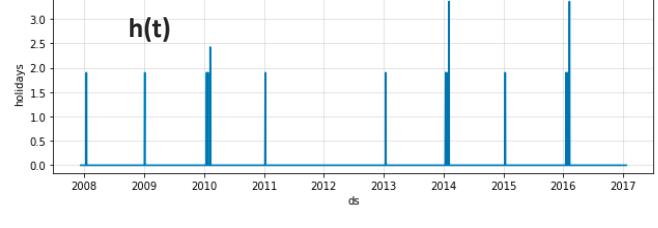
Learn More

https://medium.com/eni-digitalks/explain-me-like-im-5-how-facebook-prophet-works-and-how-to-tune-it-2183e1cf4776

Prophet's output







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Prophet Trend

Trend

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Linear with changepoints

$$g(t) = (k + \mathbf{a}(t)^\intercal \boldsymbol{\delta})t + (m + \mathbf{a}(t)^\intercal \boldsymbol{\gamma}),$$

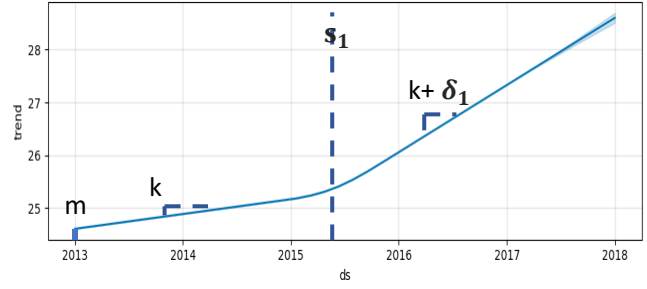
k = base growth rate (cost)

m = offset

$\boldsymbol{\gamma_j}$ = offset to make the function continuous (initialized to $-\boldsymbol{\delta_j s_j}$)

$$a_j(t) = \begin{cases} 1, & \text{if } t \geq s_j, \\ 0, & \text{otherwise.} \end{cases}$$

Logistic (saturated with capacity limit = C)



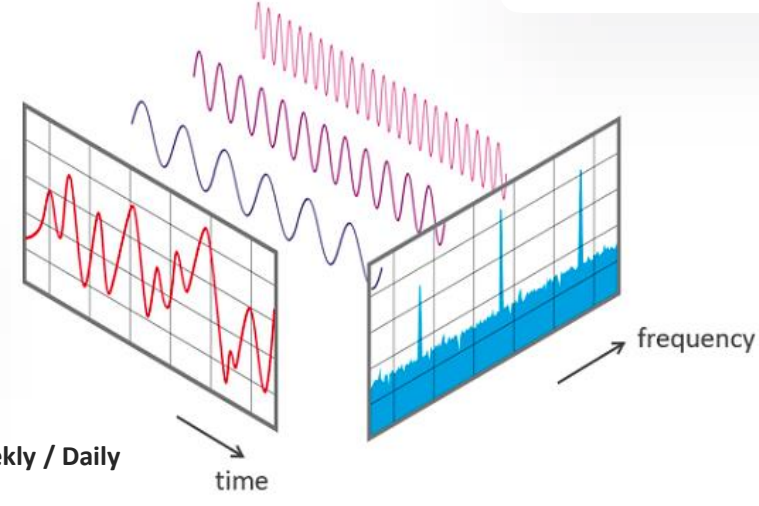
S change points to be defined
[s_1, ...,s_S]
And the variations in the angular coefficient
[δ_1, ...,δ_S]

Prophet Seasonality

Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Seasonality

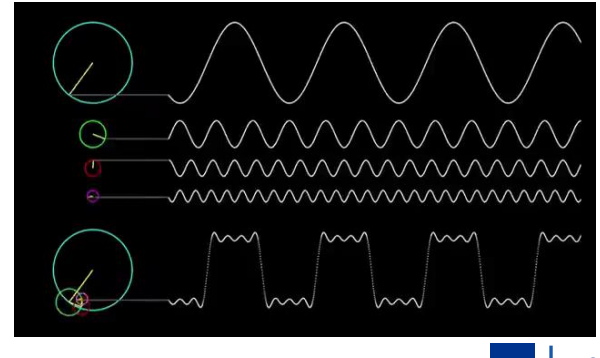


Yearly / Monthly / Weekly / Daily

Fourier (1807): any periodic function is a sum of sinusoids

Nice visualization

https://youtu.be/ds0cmAV-Yek
https://www.youtube.com/watch?v=r6sGWTCMz2k&t=432s



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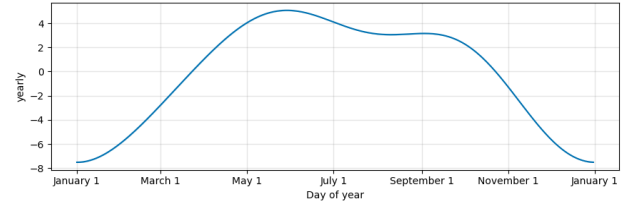
Prophet Seasonality

Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

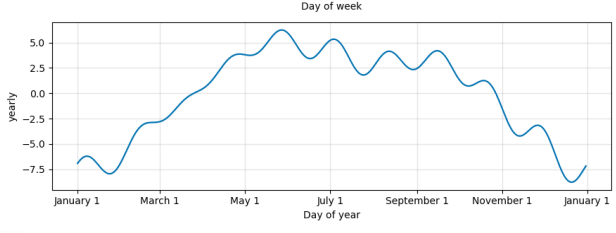
seasonality_prior_scale = 1

Fourier order = 10



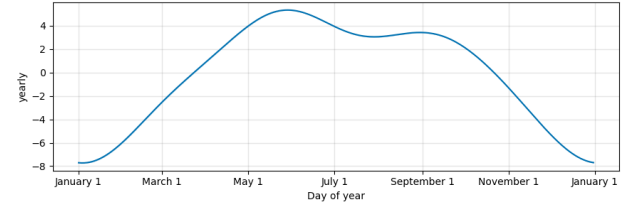
seasonality_prior_scale = 10

Fourier order = 10



seasonality_prior_scale = 10

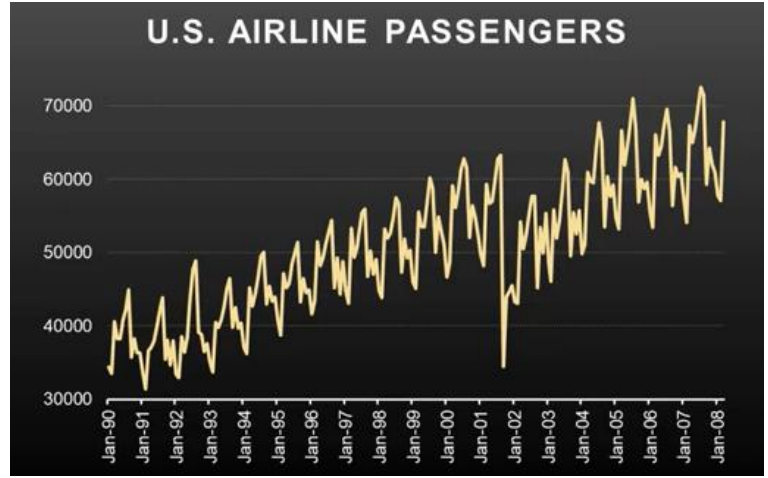
Fourier order = 2



Classic component estimation

Additive Model

1. Calculate the trend g(t) on the series
2. Return the detrended series $y(t) - g(t)$
3. Calculate the seasonal contributions as averages of the detrended series for each of the m periods: we obtain $s(t)$.
4. The residuals are calculated as $y(t) - g(t) - s(t)$

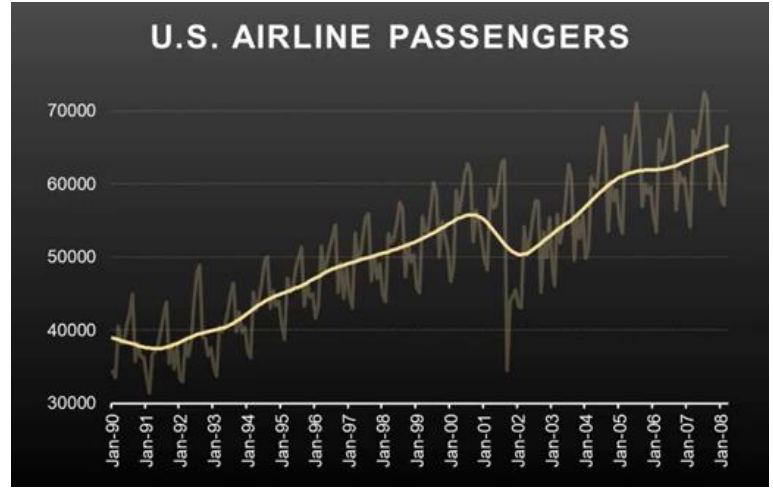


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Classic component estimation

Additive Model

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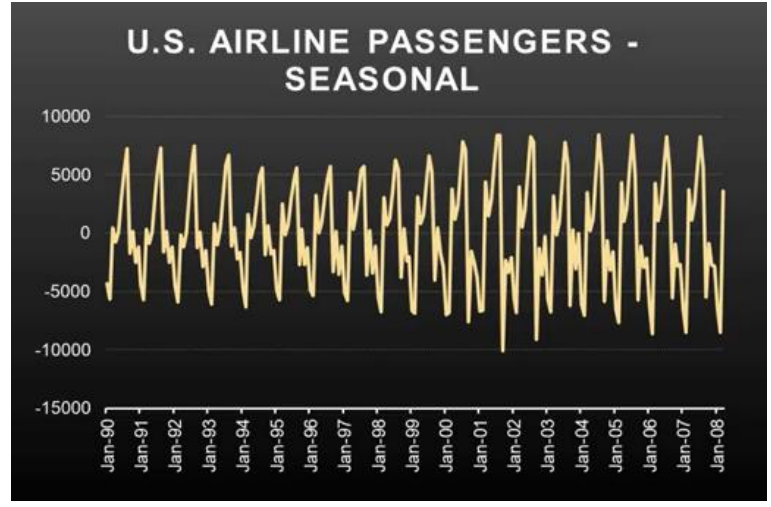


The easiest technique to calculate the trend is to make the **moving average**
There are more advanced techniques such as **LOESS Regression** – Locally estimated scatterplot smoothing

Classic component estimation

Additive Model

1. Calculate the trend g(t) on the series
2. Return the detrended series $y(t) - g(t)$
3. Calculate the seasonal contributions as averages of the detrended series for each of the m periods: we obtain $s(t)$.
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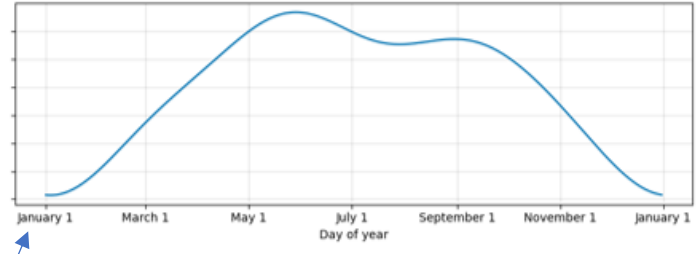


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Classic component estimation

Additive Model

1. Calculate the trend g(t) on the series
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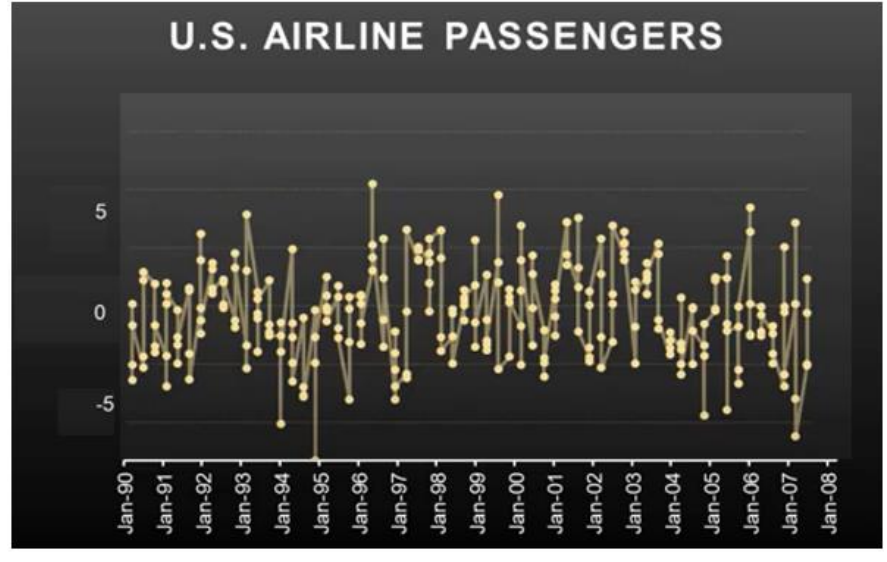


Example: the January effect is the average of all January values of the detrended series

Classic component estimation

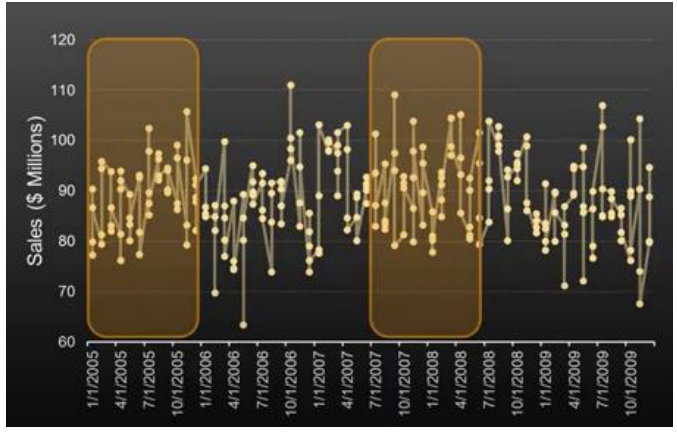
Additive Model

1. Calculate the trend g(t) on the series
2. Return the detrended series $y(t) - g(t)$
3. Calculate the seasonal contributions as averages of the detrended series for each of the m periods: we obtain $s(t)$.
4. The residuals are calculated as $y(t) - g(t) - s(t)$



What is not predicted by the sum of trend + seasonality is called error

Stationary series



A series is said to be stationary if within **two intervals** of the **same size** we find a distribution with the **same mean** and the **same variance**

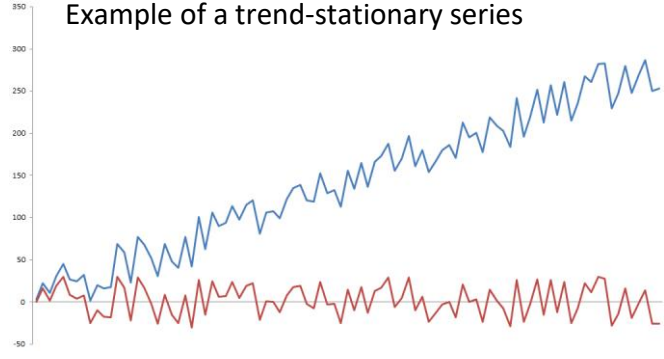
Required by ARIMA-type models; decomposition and Prophet do not need it.

Stationary series

A series can become stationary if

- Differentiated
- Trend-adjusted

Example of a trend-stationary series

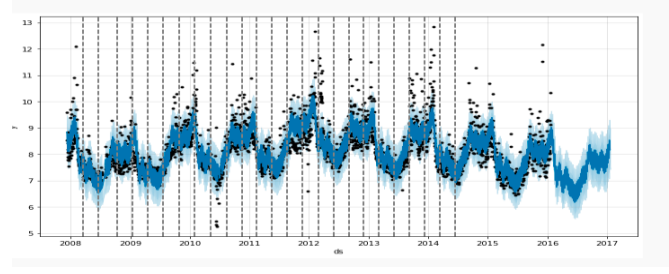


One of the ways to identify non-stationary time series is to graph the autocorrelation function (ACF). For a stationary time series, the autocorrelation function will tend to zero rather quickly, while the autocorrelation function of non-stationary data decreases slowly

Prophet Trend

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Trend



Place the S changepoints

Manually based on domain knowledge, or

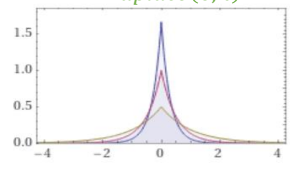
Automatically: A high number of potential equidistant changepoints (n_changepoint) are placed in the top 80% of samples (changepoint_range)

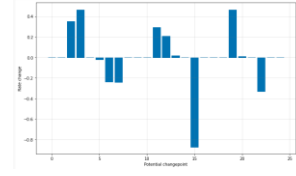
Initialize adjustments

A sparse prior is attributed to the adjustments $\boldsymbol{\delta_j \sim Laplace(0, \tau)}$ with $\boldsymbol{\tau}$ the estimated variance.

If τ is close to zero → hard trend, if τ increases → more non-zero changepoints.

$Laplace(0, \tau)$





$\tau = 0.3$ $\tau = 0.5$ $\tau = 1$

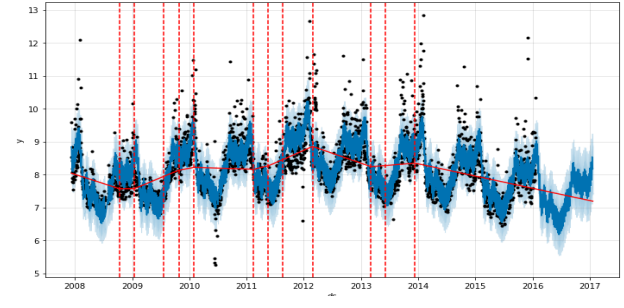
...(fit)...

Generating future trends

It is assumed that in the future there are the same number of changepoints as a percentage of the samples, so **S** changepoints every **T**
Changepoints will have adjustments distributed as **Laplace(0,"λ")**

$$\lambda = \frac{1}{S}\sum_{j=1}^{S} |\delta_j|$$

n_changepoint, changepoint_range, τ
tunable hyperparameter



Prophet Seasonality

Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Seasonality

Fourier series sum (of order N → 2N parameters: β⃗)

$$s(t) = \sum_{n=1}^{N} \left(a_n \cos\left(\frac{2\pi nt}{P}\right) + b_n \sin\left(\frac{2\pi nt}{P}\right) \right)$$

$$X(t) = \left[\cos\left(\frac{2\pi(1)t}{P}\right), \ldots, \sin\left(\frac{2\pi(N)t}{P}\right) \right].$$

$$s(t) = X(t)\boldsymbol{\beta}. \quad \boldsymbol{\beta} \sim \text{Normal}(0, \sigma^2) \quad \boldsymbol{\beta} = [a_1, b_1, \ldots, a_N, b_N]^\intercal.$$

With σ^2 the estimated variance of the terms.

N and σ^2 tunable hyperparameter

Prophet Seasonality

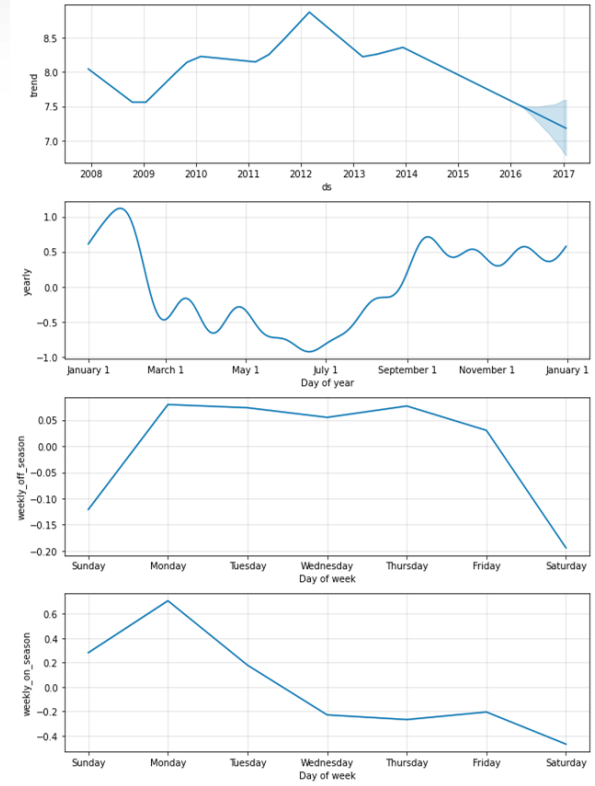
Seasonality

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

You can use custom seasonalities to differentiate effects based on external features

```python
# Python
m = Prophet(weekly_seasonality=False)
m.add_seasonality(name='weekly_on_season', period=7, fourier_order=3, condition_name='on_season')
m.add_seasonality(name='weekly_off_season', period=7, fourier_order=3, condition_name='off_season')

future['on_season'] = future['ds'].apply(is_nfl_season)
future['off_season'] = ~future['ds'].apply(is_nfl_season)
forecast = m.fit(df).predict(future)
fig = m.plot_components(forecast)
```

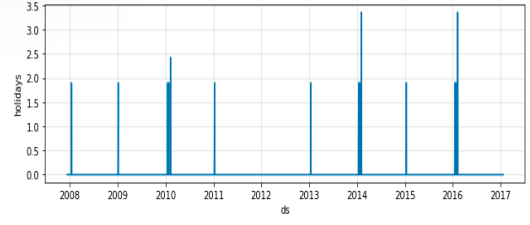


Prophet Events

Events

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

Output



Special Events

- The effects of the added events are assumed to be independent of each other
- **Automatically** added for many countries
  Or
  Manually added (0-1 dummies on dates)
- You can include upper/lower windows to lengthen the effects to neighboring dates (e.g. Christmas)

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